Alexander Black

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APPOINTMENTS

Bowdoin College — Jul 2025 –
Assistant Professor (Tenure Track)

ETH Zürich — Sep 2024 – Jul 2025
Hermann Weyl Instructor

EDUCATION

University of California at Davis, CA, USA — Sep 2019 – Jun 2024

- Ph.D. in Mathematics
  - Advised by Jesús De Loera

Cornell University, Ithaca, NY, USA — Jul 2017 – May 2019

- B.A. in Mathematics with Distinction in All Subjects
- Summa Cum Laude in Mathematics

Hamilton College, Clinton, NY, USA — Aug 2015 - Jul 2017

SELECTED PUBLICATIONS

1. *Any Proof of Polynomial Hirsch Must be Completely Incoherent* — SODA 2027
A. Black and L. Xue
arXiv:2607.11628

2. *Finding Short Paths on Simple Polytopes* — SODA 2027
A. Black and R. Steiner
arXiv:2603.05482

3. *Beyond Smoothed Analysis: Analyzing the Simplex Method by the Book* — STOC 2026
E. Bach, **A. Black**, S. Huiberts, and S. Kafer
arXiv:2510.21613

4. *Short Circuit Walks in Fixed Dimension* — SODA 2026
A. Black, C. Nöbel, and R. Steiner
arXiv:2510.01916
Invited to TALG Special Issue

5. *Small Shadows of Lattice Polytopes* — SODA 2023
A. Black
George Nicholson Paper Competition Finalist

INVITED TALKS

Beyond Smoothed Analysis: Analyzing the Simplex Method by the Book (SIAM Conference on Optimization) — Jun 2026

Non-symmetric Macdonald Polynomials and Graph Coloring (UIUC Algebra-Geometry-Combinatorics Seminar) — Apr 2026

NP-Hardness of Finding Short Paths on Polytopes (The Simplex Method: Theory, Complexity, and Applications at FU Berlin) — Jun 2025

Shadows of Polytopes (London School of Economics Seminar) — Mar 2025

Shadows of Polytopes (Polytopes and Friends at KTH Stockholm) — Dec 2024

	Exponential Lower Bounds for Many Pivot Rules for the Simplex Method (Freie Universität Berlin Discrete Geometry Seminar)	Nov 2024
	On the Shadow Simplex Method (TU Munich Seminar)	Oct 2024
	Flag Polymatroids (UC Davis Algebra and Discrete Math Seminar)	May 2024
	Monotone Paths on Polytopes in Combinatorics and Optimization (CUNEF Seminar)	Jan 2024
	Monotone Paths on Polytopes in Combinatorics and Optimization (UCLA Combinatorics Forum)	Oct 2023
	Strict Monotone Diameters of Lattice Polytopes (SIAM Conference on Optimization)	Jun 2023
	On the Circuit Diameter Conjecture for Hirsch Counterexamples (Circuit Diameters and Augmentation at the University of Colorado Denver)	May 2023
	Random Shadows of Fixed Polytopes (Institue for Computational and Experimental Research Methods (ICERM) at Brown Unversity)	May 2023
	The Polyhedral Geometry of Pivot Rules (University of Washington Seattle Combinatorics Seminar)	Apr 2023
	Smooth Torus Orbit Closures in Flag Varieties (Otto-von-Guericke-University (OVGU) Magdeburg Algebra Seminar)	Dec 2022
	Realizable Standard Young Tableaux (Freie Universität Berlin Discrete Geometry Seminar)	Dec 2022
	Small Shadows of Lattice Polytopes (Bocconi University Seminar)	Oct 2022
	Small Shadows of Lattice Polytopes (INFORMS Annual Meeting)	Oct 2022
	Torus Orbits in Full Flag Varieties (Max Planck Institute at Leipzig Nonlinear Algebra Seminar)	Sep 2022
	Small Shadows of Lattice Polytopes (Technische Universität (TU) Munich Optimization Seminar)	Sep 2022
	Small Shadows of Lattice Polytopes (ETH Zurich Computer Science Seminar)	Sep 2022
	Monotone Paths on Polytopes (University of Nebraska Lincoln Discrete Math Seminar)	Feb 2022
	Small Shadows of 0/1 Polytopes (University of Colorado at Denver Network Flows Seminar)	Jan 2022
	Polyhedral Geometry of Pivot Rules (Goethe-Universität Frankfurt-Bochum Joint Combinatorics Seminar)	Oct 2021
	Modifications of the Shadow Vertex Pivot Rule (Hausdorff Institute of Mathematics at Bonn - Trimester on Geometry of Linear Programming)	Sep 2021
CONTRIBUTED TALKS	Beyond Smoothed Analysis: Analyzing the Simplex Method by the Book (The 2026 Symposium on Theory of Computing (STOC))	Jun 2026

Short Circuit Walks in Fixed Dimension (The 2026 Symposium on Discrete Algorithms (SODA)) Jan 2026

Circuit Augmentation in Fixed Dimension (Cargese Workshop on Combinatorial Optimization) Sep 2025

Exponential Lower Bounds for Many Pivot Rules for the Simplex Method Jun 2025
(26th Conference on Integer Programming and Combinatorial Optimization (IPCO 2025))

Exponential Lower Bounds for Many Pivot Rules for the Simplex Method Nov 2024
(Oberwolfach Workshop on Combinatorial Optimization)

Exponential Lower Bounds for Many Pivot Rules for the Simplex Method (FRICO 2024) Sep 2024

Random Shadows of Fixed Polytopes (Cargese Workshop on Combinatorial Optimization) Sep 2024

The Polyhedral Geometry of Pivot Rules (Santander Workshop on Geometric Jan 2024
and Algebraic Combinatorics)

Small Shadows of Lattice Polytopes (Symposium on Discrete Algorithms (SODA23)) Jan 2023

Monotone Path Polytopes (Freie Universität Berlin Student-run Combinatorics Seminar) Dec 2022

Small Shadows of Lattice Polytopes (Cargese Workshop on Combinatorial Optimization) Sep 2022

Flag Polymatroids (Geometry meets Combinatorics in Bielefeld) Sep 2022

Monotone Paths on Cross-Polytopes (Max Planck Institute at Leipzig - (Polytop)ics: Apr 2021
Recent advances on polytopes)

Monotone Paths on Polyhedral Unit Balls (UC Davis Student Research Seminar) Feb 2021

AWARDS & SCHOLARSHIPS

- NSERC Postdoctoral Fellowship (Declined) Feb 2024
  Scholarship funds of $90,000 CAD for two years of postdoctoral study in mathematics
- 2023 Alice Siu-Fun Leung Scholarship in Mathematics Jun 2023
  $5,000 Departmental award given annually to a math graduate student at UC Davis for excellence in research
- 2022 George Nicholson Paper Competition Finalist Sep 2022
  Top 6 of 129 students in competition for the best graduate student operations research paper among students graduating after June 2021 for my paper “Small Shadows of Lattice Polytopes”
- Mixed Integer Programming (MIP) 2022 Workshop Best Poster Award May 2022
  Best poster among 30 graduate students presenting at MIP 2022
- Yueh-Jing Lin Fund Scholarship Jun 2021
  $2,000 award given to high-achieving UC Davis math graduate students
- NSF Graduate Research Fellowship Mar 2021
  Scholarship funds amounting to more than $100,000 to pursue graduate study in applied mathematics

SERVICE

- Advising Senior Thesis of Richard Lim (Joint with Jeova Farias)
- Organizer for the 2025 Mixed Integer Programming Workshop European Edition
- Poster Competition Judge at Integer Programming and Combinatorial Optimization (IPCO) 2025
- Refereed **30 papers** in total for the following venues:
  - •Innovations in Theoretical Computer Science (ITCS) [1]
  - •European Journal of Combinatorics [1]
  - •Mathematical Foundations of Computer Science (MFCS) [1]
  - •Foundations of Computer Science (FOCS) [1]
  - •Linear Algebra and its Applications [1]

- Electronic Journal of Combinatorics [1]
- Symposium on Theoretical Aspects of Computer Science (STACS) [1]
- International Mathematics Research Notices [1]
- Journal of Optimization Theory and Applications [1]
- Annals of Combinatorics [1]
- Mathematical Programming [3]
- TheoretiCS [1]
- the Symposium on Discrete Algorithms (SODA) [2]
- Combinatorial Theory [1]
- the SIAM Journal of Discrete Mathematics (SIDMA) [3]
- Integer Programming and Combinatorial Optimization (IPCO) [3]
- Operations Research Letters [2]
- Mathematics of Operations Research [1]
- Symposium on the Theory of Computing (STOC) [1]
- Discrete and Computational Geometry [1]
- Formal Power Series in Algebraic Combinatorics (FPSAC) [1]
- Bulletin of the London Mathematical Society [1]

- Volunteered to help set up the 2023 conference on Formal Power Series and Algebraic Combinatorics (FPSAC)
- Jointly ran a preparation workshop for the GRE Quantitative section for the University of Nebraska Lincoln McNair Scholars program in summer 2023
- Co-founded and ran a commutative algebra REU for first generation, low-income students at University of Nebraska Lincoln in summer 2023 (PI: Eloísa Grifo) that has been funded and graduate student run annually again in summers 2024 and 2025
- Co-organizer of the student/postdoc seminar at ICERM at Brown University for the Spring 2023 semester on discrete optimization
- Co-founder of the student-run algebra and discrete math seminar at UC Davis
- Co-advised research of undergraduate Owen Gao at UC Davis